

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool : Data Interpretation

Weightage: 20%

QUESTIONS:

Total Marks: 20

1: HTTP Response Analysis

A web server returns the following HTTP response header:

```
HTTP/1.1 200 OK
Date: Mon, 01 Apr 2026 10:00:00 GMT
Content-Type: text/html
Content-Length: -150
Connection: keep-alive
```

Questions:

1. Identify any incorrect or invalid fields in the header.
2. Analyze the issue with `Content-Length` and explain its impact.
3. Determine whether the status code matches the response correctness.
4. Suggest corrections for the identified errors.

2: DOM Structure Analysis

Consider the following DOM tree representation:

```
<html>
  <head>
    <title>Test Page</title>
  </head>
  <body>
    <div>
```

```
<p>Hello World
</div>
</body>
</html>
```

Questions:

1. Identify missing or improperly closed elements.
2. Analyze how the browser will interpret this structure.
3. Determine the impact on rendering and layout.
4. Suggest corrections to make the DOM valid.

3: GET vs POST in Login System

A login system uses two different methods:

Method	URL Example	Data Visibility	Security Level
GET	/login?user=admin&pass=1234	Visible in URL	Low
POST	/login	Hidden in body	Moderate

Questions:

1. Interpret the differences in data transmission between GET and POST.
2. Analyze which method is more suitable for login and why.
3. Identify security risks in using GET for authentication.
4. Suggest best practices for secure login implementation.

4: HTML Code Inspection

Analyze the following HTML snippet:

```
<!DOCTYPE html>
<html>
<head>
<title>Sample</title>
</head>
<body>
<h1>Welcome
<p>This is a paragraph

</body>
</html>
```

Questions:

1. Identify missing closing tags.
2. Analyze issues with the `<img>` tag.
3. Determine how browsers will handle these errors.
4. Suggest a corrected version of the code.

5: Browser-Server Communication Flow

The following sequence describes a web request:

1. User enters a URL in the browser
2. DNS lookup occurs

3. HTTP request is sent
4. Server processes request
5. Response is returned
6. Browser renders page

Questions:

1. Interpret each step in the communication flow.
2. Identify where delays can occur.
3. Analyze the role of DNS in the process.
4. Suggest optimizations to improve performance.

Code of Conduct:

*I **E.HEMANTH REDDY(192372098)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: E.HEMANTH REDDY

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided

Application of Concepts	20%	Effectively applies relevant concepts/ tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

LOGIN

STUDENT ACCESS PORTAL

SIMATS ACADEMIC PORTAL

Internet Programming (CSA43) - Course Outcome CO-1 Assessment

 **SIMATS Web Portal**

LOGIN

HOME

HTTP & LOGIN

HTML & DOM

BROWSER FLOW

 **Student Login**

Please enter your Student Name and Register Number to access the assessment portal. Your details will be displayed at the bottom-right corner of all pages.

Student Full Name:

Register Number / Roll No:

Login & Access Portal →

INTERFACE:

 **SIMATS Web Portal**

LOGIN

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BROWSER FLOW

 **Course & Assessment Information**

CO-1 Assessment

Parameter	Details
Course Code & Title	CSA43 - Internet Programming
Course Outcome Covered	CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)
Assessment Tool	Data Interpretation (Weightage: 20% Total Marks: 20)
Target Evaluation Level	Level 4 (Exemplary Solution)

 **Assessment Modules**

5 Questions

Module Page	Questions Addressed	Key Web Standards Concepts	Action
HTTP & Login Security	Question 1 & Question 3	RFC 9110 Content-Length validation, status code 200 OK mismatch, GET vs POST transmission security.	Open Page
HTML & DOM Inspection	Question 2 & Question 4	HTML5 DOM tree auto-closure, unclosed tags, WCAG image accessibility, and CLS optimization.	Open Page
Browser Communication	Question 5	6-step web request flow, DNS IP resolution, latency bottlenecks, and network optimizations.	Open Page

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QUESTION 1:

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HTML & DOM

BROWSER FLOW

1: HTTP Response Analysis (Question 1)

4 Marks

A web server returns the following HTTP response header:

HTTP/1.1 200 OK Date: Mon, 01 Apr 2026 10:00:00 GMT Content-Type: text/html Content-Length: -150 Connection: keep-alive

Header Diagnostics & Analysis Table

Header Field	Received Value	Status	RFC Specification & Impact Analysis
Content-Length	-150	Invalid	According to RFC 9110 58.6, Content-Length MUST be a non-negative decimal integer. A negative length causes parser exceptions, stream truncation, connection hanging, or socket buffer errors.
Status Code	200 OK	Mismatch	Returning 200 OK with a malformed header is improper. If the server encountered an error calculating length, it should return 500 Internal Server Error.
Content-Type	text/html	Valid	Valid MIME type, though standard practices require appending character set metadata (e.g. charset=UTF-8).

Corrected HTTP Response Header

HTTP/1.1 200 OK Date: Mon, 01 Apr 2026 10:00:00 GMT Content-Type: text/html; charset=UTF-8 Content-Length: 150 Connection: keep-alive

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QUESTION 2:

DOM STRUCTURE & HTML INSPECTION

Data Interpretation Solutions for HTML5 DOM Parsing, Unclosed Tags, and Accessibility Validation

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2: DOM Structure Analysis (Question 2)

4 Marks

Consider the following DOM tree representation:

<html> <head> <title>Test Page</title> </head> <body> <div> <p>Hello World</p> </div> </body> </html>

DOM Analysis & Auto-Correction Breakdown

Analysis Item	Observed Defect	Browser Interpretation Mechanism	Impact on Layout & Rendering
Paragraph Tag	Missing </p> tag	HTML5 tree construction algorithm auto-closes the paragraph when encountering the closing block container </div>.	May corrupt child selectors (e.g. div > p) or layout rendering if subsequent elements follow.
Document Type	Missing <!DOCTYPE html>	Browser defaults to legacy Quirks Mode rendering.	Causes non-standard box model rendering and cross-browser visual layout shifts.

Corrected Standards-Compliant DOM Code

<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <title>Test Page</title> </head> <body> <div> <p>Hello World</p> </div> </body> </html>

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QUESTION 3:

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LOGIN

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HTTP & LOGIN

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BROWSER FLOW

3: GET vs POST in Login System (Question 3)

4 Marks

A login system uses two different data transmission methods:

Method	URL Example	Data Visibility	Security Level
GET	/login?user=admin&pass=1234	Visible in URL Bar	Low (Vulnerable)
POST	/login	Hidden in HTTP Body Payload	Moderate / Secure (with HTTPS)

Detailed Security Comparison

Security Aspect	GET Method Impact	POST Method Impact
Browser History	Credentials remain stored cleartext in browser history.	Data excluded from browser history.
Server Access Logs	Logged plain text in Nginx/Apache access.log.	Credentials hidden from access log files.
Referrer Header Leaks	Transmitted to external links via Referrer header.	Protected against Referrer header leaks.
Authentication Suitability	✗ Unsuitable & Vulnerable	✓ Standard & Mandatory Method

Best Practices for Secure Login

- Submit authentication requests exclusively via HTTP POST.
- Enforce HTTPS / TLS 1.3 Encryption to secure transport layer traffic.
- Use Anti-CSRF Tokens and set sameSite=strict on session cookies.
- Hash stored passwords server-side using Argon2id or bcrypt with unique salts.

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QUESTION 4:

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LOGINHOMEHTTP & LOGINTHTML & DOMBROWSER FLOW

4: HTML Code Inspection (Question 4)

4 Marks

Analyze the following raw HTML snippet:

<!DOCTYPE html> <html> <head> <title>Sample</title> </head> <body> <h1>Welcome <p>This is a paragraph </body> </html>

Inspection Findings & Errors Table

Element Tag	Identified Defect	Standards Violation	Browser Recovery Action
<h1>	Missing closing </h1> tag	Unclosed heading block.	Parser auto-closes <h1> upon encountering subsequent block tag <p>.
<p>	Missing closing </p> tag	Unclosed paragraph block.	Parser auto-closes <p> before closing </body>.
	Missing mandatory alt attribute	WCAG 2.1 Accessibility Violation.	Renders image without text fallback for screen readers. Missing dimensions cause Cumulative Layout Shift (CLS).

Corrected Standards-Compliant HTML Snippet

<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <title>Sample</title> </head> <body> <h1>Welcome</h1> <p>This is a paragraph</p> </body> </html>

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QUESTION 5:

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LOGINHOMEHTTP & LOGINTHTML & DOMBROWSER FLOW

5: Browser Communication Lifecycle (Question 5)

4 Marks

The following sequence describes a complete web request lifecycle:

1
User Input
URL entered in browser

2
DNS Lookup
Hostname resolved to IP

3
HTTP Request
TCP/TLS handshake sent

4
Server Process
Backend builds payload

5
HTTP Response
Status 200 & stream

6
Render Page
DOM/CSSOM & Painting

Detailed Lifecycle Step Interpretation

1. **User enters URL:** Browser parses protocol ('https://'), domain hostname, and target path.

2. **DNS lookup:** Browser queries recursive resolvers to translate domain name into IP address ('193.0.2.1').

3. **HTTP request sent:** Initiates TCP 3-Way Handshake ('SYN', 'SYN-ACK', 'ACK') & TLS negotiation, sending HTTP GET headers.

4. **Server processes request:** Web server routes request to application logic, queries database, constructs HTML stream.

5. **Response returned:** Server returns status line ('200 OK'), response headers, and content stream.

6. **Browser renders pages:** Critical Rendering Path tokenizes HTML $\rightarrow$ DOM Tree; CSS $\rightarrow$ CSSOM Tree; combines $\rightarrow$ Render Tree $\rightarrow$ Paint.

Latency Bottlenecks & Optimizations

Analysis

1. Delay Points Table

Lifecycle Stage	Bottleneck Type	Detailed Latency Rationale
Step 2 (DNS Lookup)	DNS Resolution Delay	Uncached DNS queries require recursive lookups across root, TLD, and authoritative nameservers, adding 50-300ms network RTT.

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Latency Bottlenecks & Optimizations

Analysis

1. Delay Points Table

Lifecycle Stage	Bottleneck Type	Detailed Latency Rationale
Step 2 (DNS Lookup)	DNS Resolution Delay	Uncached DNS queries require recursive lookups across root, TLD, and authoritative nameservers, adding 50–300ms network RTT.
Step 3 (HTTP Request)	TCP / TLS Handshake Overhead	Establishing TCP socket connections and negotiating TLS 1.3 encryption keys requires multiple RTT round-trips over distance.
Step 4 (Server Processing)	Time to First Byte (TTFB)	Unoptimized backend database queries or heavy application logic delay initial response streaming.
Step 6 (Browser Rendering)	Render-Blocking Resources	Synchronous external <script> tags and large CSS stylesheets pause DOM construction.

2. Performance Optimizations Table

Optimization Technique	Implementation Mechanism	Performance Impact
DNS Prefetching & Preconnect	<link rel="dns-prefetch" href="...">	Resolves DNS hostnames in advance, eliminating resolution latency.
Content Delivery Network (CDN)	Edge Caching Servers	Serves static assets from edge locations physically closer to users, reducing RTT.
HTTP/2 & HTTP/3 Protocols	Connection Multiplexing	Eliminates Head-of-Line blocking by streaming multiple requests over a single TCP/QUIC connection.

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RESULT: WE Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery.